Maxima and Minima

Mind map

Maxima / minima at a point
- $f(a) > f(a \pm h)$: max
- $f(a) < f(a \pm h)$: min
- Continuity not needed

Boundary points
- Compare with the side in domain
- Points of the domain only

Synonyms
- Local = relative
- Greatest = global = absolute
- Extrema = turning points

Greatest / least value
- Critical points + end points
- Max value may be < min value

First derivative test
- $+ \to -$: max; $- \to +$: min
- No sign change: neither
- Also where f' not defined

Second derivative test
- $f''(a) > 0$: min
- $f''(a) < 0$: max
- $f''(a) = 0$: fails

nth derivative test
- n even: extremum
- n odd: no extremum

Concavity
- $f'' > 0$: up, $f' \uparrow$
- $f'' < 0$: down, $f' \downarrow$
- Tangent below / above

Inequalities by concavity
- Chord above curve if $f'' > 0$
- $\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}$

Point of inflection
- Concavity changes
- $f'' = 0$ / undefined + sign change
- Extremum of f'

Geometrical problems
- Figure, one variable
- $f' = 0$ + end values
- Mensuration formulas

Cubic: number of roots
- $f(\alpha)f(\beta) < 0$: 3 roots
- > 0 or $D \leq 0$: 1 root

Cubic: sign of extrema
- $a > 0$: max at smaller root
- Location of roots of $f' = 0$
- Inflection $x = -\frac{b}{3a}$

Polynomial from extrema
- $f' = k(x - \alpha)(x - \beta)$
- Limit kills low terms